

Microeconomic Foundations, I: Choice and Competitive Markets

Instructor's Manual

Chapter 13: Aggregating Firms and Consumers

Notes on the Chapter

This chapter does two fairly obvious things and one that is perhaps a bit less obvious. On the obvious side is the straightforward result that a finite collection of profit-maximizing firms aggregate into one profit-maximizing “mega” firm, whose production-possibility set is the Minkowski sum of the production-possibility sets of the original firms and whose (resulting) profit function is just the sum of the profit functions of the firms.

And then, for consumers, the “usual suspects” in results are given: Identical and homothetic consumers can be aggregated without worrying about the distribution of income; if you fix the distribution of income, you can drop “identical” if you keep “homothetic”; but (then), if you move from demand to excess demand (where each consumer’s wealth is given by the market value of her endowment), the Sonnenschein–Mantel–Debreu (S-M-D) Theorem applies: All you can say about aggregate excess demand (as a function) is that it is continuous, homogeneous of degree zero, and satisfies Walras’ Law, at least for prices that are uniformly bounded away from the boundary (when normalized to have norm 1).

I do not give a proof of S–M–D. I give the statement of the theorem as presented by Debreu (1974), and in Problem 13.6 (solution in this *Manual*) I provide the excess demand functions of the consumers in Debreu’s proof, inviting the student to prove that these functions are continuous and satisfy Walras’ Law and the weak axiom of revealed preference and sum to the initially given aggregate excess demand. But the proof that these are the excess demand functions of utility-maximizing consumers is too much; the interested student will have to consult Debreu (1974) or Geanakoplos (1984), in which the utility functions are identified.

The less than obvious part of the chapter concerns convexification through aggregation. After a simple example to motivate the idea, I grind through the proof of the Shapley–Folkman Theorem (using Starr’s notion of the inner radius of a set) and then (a bit informally) show how this applies to yield “nearly convex” *per capita* excess demand, when the number of consumers is much larger than the number of commodities.

Finally, I discuss (only discuss!) the idea of having a continuum of consumers (or firms) and give (informally) the result that, for nonatomic measures of agents, the Aumann integral of excess demand sets is immediately convex. I emphasize, this is only loose discussion; students with a nodding acquaintance may be able to read this and get the basic idea, but there is no sense in which anything of substance is proved.

As I wrote the chapter, I thought of closing with a discussion of how aggregation of a suitably diverse but large group of (say) consumers would turn excess demand correspondences into aggregate (per capita) excess demand *almost-functions*. But I wasn't able to find a good way to explain the "diversity" (that is, genericity) issues without moving into differential geometry/topology. Not wishing to go down that path, I gave up on this. (You can anticipate that similar concerns will arise next chapter, when it comes to discussing things like the generic finiteness of the set of Walrasian equilibria.)

This chapter was written late in the process of writing the book, to fill an obvious hole in coverage. As a result, as I finish the final draft of the book, I have taught the material in this chapter only once. In that class session, I covered both Chapters 12 and 13, and more of the ninety minutes went to developments of Chapter 12 than to this chapter. I emphasized what the questions were and why they were important, the idea that things worked beautifully for firms but not for consumers, and then what S-M-D was saying. I said nothing about convexification through aggregation. So I'm not much help on how to cover this material. If anyone has any good ideas on the subject, please email me; perhaps I will do better in later versions.

Notes on the Problems

My lack of experience with this chapter extends to the problems. I have no useful insights to offer.

Solutions to Non-Starred Problems

■ 13.2. (a) If each individual demand function is homogeneous of degree zero in prices and income, it is obvious that so is the sum of those functions, or aggregate demand. And it is nearly as obvious that if each individual demand function satisfies Walras' Law, then so does aggregate demand. But just to write this bit out, if $p \cdot d^h(p, y^h) = y^h$ for each h , then

$$p \cdot \left[\sum_h d^h(p, y^h) \right] = \sum_h p \cdot d^h(p, y^h) = \sum_h y^h = y.$$

The (only slightly) hard part is showing that the Slutsky substitution matrix for aggregate demand is symmetric and negative semi-definite, if each individual-demand Slutsky matrix has these properties. But this is only slightly hard: First note that the

sum of symmetric and negative semi-definite matrices is symmetric and negative semi-definite. So we have the desired result once we show that the Slutsky matrix for aggregate demand is the sum of the Slutsky matrices for the individual demand functions.

And, sure enough, if we write $d(p, y)$ for aggregate demand,

$$\frac{\partial d_j}{\partial p_i} + d_i \frac{\partial d_j}{\partial y} = \frac{\partial \sum_h d_j^h}{\partial p_i} + [\sum_h d_i^h] \frac{\partial d_j}{\partial y} = \sum_h \left[\frac{\partial d_j^h}{\partial p_i} + d_i^h \frac{\partial d_j}{\partial y} \right].$$

The key (to both steps in the preceding chain of equalities) is to note that if condition (13.1) holds, then $\partial d_j / \partial y = \partial d_j^h / \partial y = b_j(p)$, for all h .

(b) Suppose indirect utility for consumer h has the form $\nu^h(p, y^h) = \alpha^h(p) + \beta(p)y^h$. Assuming all derivatives that are taken can be taken, Roy's Identity tells us that

$$d_i^h(p, y) = - \left(\frac{\partial \nu^h}{\partial p_i} \right) / \left(\frac{\partial \nu^h}{\partial y} \right) = - \left(\frac{\partial \alpha^h}{\partial p_i} + y^h \frac{\partial \beta}{\partial p_i} \right) / \beta(p).$$

This is the form (13.1), where

$$a_i^h(p) = - \frac{1}{\beta(p)} \frac{\partial \alpha^h}{\partial p_i} \quad \text{and} \quad b_i(p) = - \frac{1}{\beta(p)} \frac{\partial \beta}{\partial p_i}.$$

■ 13.3. Take consumer 1 first. At prices (1, 2), the consumer demands (7, 6.5), which costs 20. The cost of the "other" bundle (7, 6) at these prices is 19, so we know (by the logic of Afriat's Theorem) that this consumer must have $(7, 6.5) \succ (7, 6)$. (This is hardly surprising, since the bundles have the same amount of good 1 and the first has more good 2.) And at prices (2, 1), bundle (7, 6) costs 20 while (7, 6.5) costs 20.5. So this consumer is okay; she cannot afford the strictly preferred (7, 6.5).

For consumer 2, everything is symmetrical. At the prices (2, 1), he chooses (6.5, 7) which costs 20, while (6, 7) would cost only 19. So he must strictly prefer the first bundle. And at prices (1, 2), he chooses (6, 7), which costs 20, while the preferred (6.5, 7) is unaffordable at 20.5.

But now (and this is just repeating from the text), at prices (1, 2) the aggregate demand is $(7, 6.5) + (6, 7) = (13, 13.5)$, costing 40, while the "other" aggregate demand bundle is $(7, 6) + (6.5, 7) = (13.5, 13)$, which costs 39.5. And at prices (2, 1), the chosen aggregate demand costs 40 while the unchosen bundle costs 39.5. This violates GARP (and, indeed, is a violation of WARP).

■ 13.6. Excess demand $\zeta^h(p)$ is being defined on Q^ϵ , so homogeneity is no problem: we extend it so it is homogeneous of degree zero (on all price vectors p such that $p/\|p\| \in Q^\epsilon$).

Continuity is no problem; $\zeta^h(p)$ is made up of a elementary (continuous) operations performed on continuous functions.

Concerning Walras' Law, in computing $p \cdot \zeta^h(p)$, the scalar $\zeta_h(p) + \alpha(p)p_h$ is irrelevant; the question is, What is $p \cdot (e^h - p_h p)$? And it is $p \cdot e^h - p \cdot (p_h p) = p_h - p_h(p \cdot p) = 0$, since $p \cdot p = \|p\|^2 = 1$.

Next, we compute $\sum_h (\zeta_h(p) + \alpha(p)p_h)(e^h - p_h p)$; we want to show that this is $\zeta(p)$. Write the summation as

$$\begin{aligned} \sum_h (\zeta_h(p) + \alpha(p)p_h)e^h - \sum_h (\zeta_h(p) + \alpha(p)p_h)p_h p = \\ \sum_h (\zeta_h(p) + \alpha(p)p_h)e^h - \sum_h \zeta_h(p)p_h p - \sum_h \alpha(p)(p_h)^2 p. \end{aligned}$$

Recall that $H = k$; there are as many consumers as there are commodities. Then we can take each of the three summations on the right-hand side of the display in turn:

Concerning the first summation, since the vector e^h is 1 in coordinate position h and 0 in all other coordinate positions, the term $(\zeta_h(p) + \alpha(p)p_h)e^h$ is 0 in position $h' \neq h$ and $(\zeta_h(p) + \alpha(p)p_h)$ in position h . So the first sum in right-hand side of the previous display is the vector whose h' th coordinate is $\zeta_{h'}(p) + \alpha(p)p_{h'}$.

Concerning the second summation, in coordinate position h' , this is just $p_{h'} \sum_h \zeta_h(p)p_h = p_{h'}(p \cdot \zeta(p)) = 0$, since ζ satisfies Walras' Law.

And concerning the third summation, in coordinate position h' , this is $p_{h'} \alpha(p) \sum_h (p_h)^2 = p_{h'} \alpha(p)$.

So in coordinate position h' , the right-hand side of the previous display is

$$\zeta_{h'}(p) + \alpha(p)p_{h'} - \alpha(p)p_{h'} = \zeta_{h'}(p),$$

which is just what we wanted to show.

And, looking at the second two summations on the right-hand side of the display together, the h' th coordinate is

$$\sum_h \zeta_h(p)p_h p_{h'} - p_{h'} \sum_h \alpha(p)(p_h)^2.$$

Since $p \cdot \zeta(p) = 0$ by Walras law, and

$$p_{h'} \sum_h \alpha(p)(p_h)^2 = \alpha(p)p_{h'} \sum_h p_h^2 = \alpha(p)p_{h'}.$$

So, combining the terms, we get coordinate position h' of $\sum_h \zeta^h(p)$ is

$$\zeta_{h'}(p) + \alpha(p)p_{h'} - 0 - \alpha(p)p_{h'} = \zeta_{h'}(p).$$

The final task is to verify that each ζ^h satisfies the Weak Axiom of Revealed Preference. We suppose that for two distinct price vectors p and p' , $p' \cdot \zeta^h(p) \leq 0$. As in showing that Walras' Law holds, the scalar is irrelevant to the sign of $p' \cdot \zeta^h(p)$, so $p' \cdot \zeta^h(p) \leq 0$ implies $p' \cdot (e^h - p_h p) = p'_h - p_h(p' \cdot p) \leq 0$. This is $(p'_h/p_h) \leq p' \cdot p$. But because we are looking at distinct prices normalized to have norm 1, $p' \cdot p < 1$. Therefore, $p' \cdot \zeta^h(p) \leq 0$ for distinct p and p' implies $p'_h < p_h$. Of course, this means that $p' \cdot \zeta^h(p) \leq 0$ and $p \cdot \zeta^h(p') \leq 0$ cannot both be true (unless $p = p'$), since $p'_h < p_h$ and $p_h < p'_h$ cannot both be true. The Weak Axiom of Revealed Preference holds.